

Power Plays and Winning in Mixed Doubles Curling: A Bayesian Analysis

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Abstract

In Mixed Doubles curling, the Power Play is a tactical option designed to create scoring opportunities. This report explores whether using a Power Play meaningfully improves the US team’s odds of winning a match. We analyzed 311 ends from international competition from 2022 to 2025 using a Bayesian hierarchical logistic regression model. We separated different tactical effects from the team’s overall performance by including intercept. Our result tells that calling a Power Play does not directly increase the odds of winning. Scoring high point correlates with lower win odds, which likely because the US team usually attempt doing high-risk play when they are already behind. Therefore, we conclude that Power Play is a tactic to save the game when it’s already losing, while Power Play is not a tool that guarantees a win.

Background and Objective

Curling is a strategic team sport in which players attempt to place stones closer to the center of a target than their opponent. In Mixed Doubles curling, the format introduces additional strategic complexity through features such as pre-placed stones and the option to use a “power play,” which alters the positioning of these stones to create more aggressive scoring opportunities. Because each team is allowed only one power play per game, deciding when and whether to use it is a critical strategic decision.

This report is motivated by the question of whether employing a power play meaningfully impacts a team’s chances of winning. In particular, we aim to determine whether the odds of winning a match differ depending on whether a power play is used during an end, and whether this relationship depends on the effectiveness of the shot executed during that end. Shot effectiveness is measured on a scale from 0 to 4, with higher values indicating better execution. For the purposes of this analysis, we define a “successful” shot as one with a score of 3 or 4, and an “unsuccessful” shot as one with a score of 0, 1, or 2.

The dataset used in this analysis consists of observations from matches played by the United States Mixed Doubles curling team across several international competitions, including the 2022 Winter Olympics in Beijing and the World Mixed Doubles Curling Championships from 2023 to 2025. Each observation corresponds to a single end within a match and includes indicators for whether a power play was used, whether the match was ultimately won, and the effectiveness of the shot.

Preliminary data exploration suggests variation in win rates across competitions, as well as differences in shot effectiveness between power play and non-power play situations. These patterns motivate a more formal modeling approach to quantify the relationship between power plays, shot success, and match outcomes.

Methodology

We analyze the impact of Power Plays and their success on the probability that the U.S. Mixed Doubles team wins an end. Our dataset consists of $i = 1, \dots, N = 311$ ends from $m = 1, \dots, M = 41$ distinct

matches. For each end, we record whether the U.S. team won the end (USWin_{*i*}), whether a Power Play was executed (PowerPlay_{*i*}), and whether the shot was successful (Success_{*i*}). Each end is nested within a specific match (indexed by match_id_{*i*}), which allows us to account for potential differences between matches. By specifying $\alpha_{\text{match}} \sim \text{Normal}(0, \sigma_{\text{match}})$, the model incorporates the partial pooling method. This hierarchical design allows sharing information across all 41 matches. It helps stabilize our results for matches with less data, while still respecting the fact that every game has its own unique difficulty level.

Since the outcome USWin_{*i*} is binary, we modeled the probability of winning using a Bernoulli likelihood with a logit link. We included an interaction term to directly address whether the effect of a Power Play depends on its success. We also included a random intercept for each match to capture match-specific variation in baseline win probability. Formally, the model is specified as

$$\begin{aligned} \text{USWin}_i &| p_i \sim \text{Bernoulli}(p_i), \\ \text{logit}(p_i) &= \beta_0 + \beta_1 \text{PowerPlay}_i + \beta_2 \text{Success}_i + \beta_3 (\text{PowerPlay}_i \cdot \text{Success}_i) + \alpha_{\text{match}(i)}, \\ \beta_0 &\sim \text{Normal}(0.5, 1), \\ \beta_1 &\sim \text{Normal}(0, 1), \\ \beta_2 &\sim \text{Normal}(0, 1), \\ \beta_3 &\sim \text{Normal}(0, 1), \\ \alpha_{\text{match}} &\sim \text{Normal}(0, \sigma_{\text{match}}), \\ \sigma_{\text{match}} &\sim \text{Half-Normal}(0, 0.5), \end{aligned}$$

where β_0 represents the baseline log-odds of winning an end, β_1 and β_2 capture the main effects of executing a Power Play and achieving a successful shot, respectively, and β_3 models the interaction between Power Play and success. The random intercepts α_{match} allow the baseline probability of winning to vary across matches, accounting for match-specific differences such as opponent strength or game conditions.

We chose standard normal priors to stabilize estimation while allowing the data to influence the results. The intercept β_0 is assigned a $\text{Normal}(0.5, 1)$ prior, corresponding to a baseline win probability of roughly 60-65% on the probability scale ($\text{logit}^{-1}(0.5) = \frac{e^{0.5}}{1+e^{0.5}} \approx 0.62$), which aligns with the overall win rate observed in the data in the competitions. β_1 , β_2 , and β_3 have priors allowing a wide range of plausible effects while still discouraging extreme values. The prior on the standard deviation of match intercepts, σ_{match} , is Half-Normal, reflecting the expectation that match-level variability is non-negative but not unreasonably large.

The model was implemented in Stan using the rstan R package. The tidyverse and bayesplot packages were used for data wrangling and visualization. We used a Bernoulli likelihood with logit link for the binary outcome, and included match-level random intercepts to capture correlation within matches. Including the interaction term addresses the primary research question of whether the effect of executing a Power Play depends on its success. Posterior inference was obtained using four chains, each with 2000 iterations (1000 warm-up), resulting in 4000 post-warm-up draws. Convergence diagnostics ($\hat{R}$) indicated that the chains mixed well, and effective sample sizes were sufficient for all parameters.

Analysis and Results

We ran the Bayesian hierarchical logistic regression model described above, which includes a random intercept for match and an interaction term between Power Play execution and shot success. Posterior summaries for the key coefficients are presented below in Table 1 on the logit scale and converted to the odds ratio scale later in Table 2.

Parameter	Posterior Mean	95% Credible Interval	Interpretation
β_0 (Intercept)	1.08	[-0.13, 2.30]	Baseline log-odds
β_1 (Power Play)	-0.04	[-1.52, 1.43]	Power Play effect
β_2 (Success)	-0.08	[-1.25, 1.10]	Shot success effect
β_3 (Interaction)	-0.23	[-1.74, 1.35]	Interaction effect
σ_{match}	3.02	[2.48, 3.61]	Substantial match-to-match variability

Table 1: Posterior summaries for model coefficients (logit scale).

All coefficients are reported on the log-odds (logit) scale. The posterior mean for the intercept, $\beta_0 = 1.08$, represents the baseline log-odds of winning a match when no Power Play is used and the shot is unsuccessful. Converting to the probability scale, this corresponds to odds of $e^{1.08} \approx 2.94$, or a probability of about 74% $\left(\frac{2.94}{1+2.94} \approx 0.74\right)$. This suggests that even in less favorable situations, the U.S. team still has a relatively high probability of winning the match.

The posterior means and credible intervals for β_1 , β_2 , and β_3 all remain close to zero with wide intervals, indicating substantial uncertainty about the effects of executing a Power Play, a successful shot, or their interaction on the probability of winning a match. Because the credible intervals for these parameters all include 0, we cannot rule out either positive or negative effects, and there is no strong evidence that these factors meaningfully influence match outcomes.

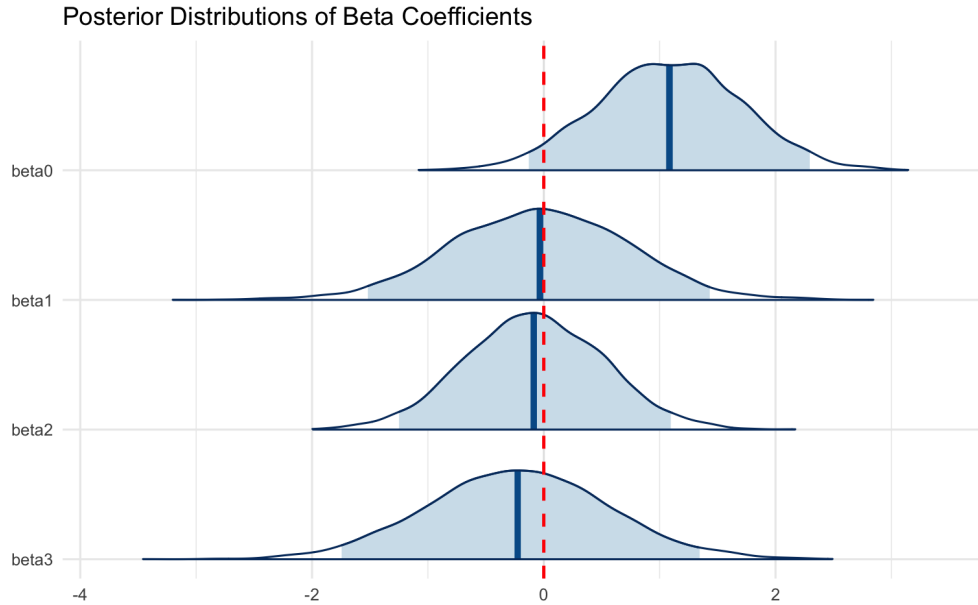


Figure 1: Density plots of posterior draws for the β coefficients (all chains merged), with uncertainty intervals shown as shaded regions. The red dashed line marks zero.

Parameter	Odds Ratio (Mean)	95% Credible Interval	Interpretation
β_0 (Intercept)	2.94	[0.88, 9.97]	Baseline odds of US winning a match
β_1 (Power Play)	0.96	[0.22, 4.18]	Multiplicative effect of Power Play
β_2 (Success)	0.92	[0.29, 3.00]	Multiplicative effect of shot success
β_3 (Interaction)	0.79	[0.18, 3.86]	Interaction effect on odds scale

Table 2: Posterior summaries for model coefficients on the odds ratio scale. Values are exponentiated from log-odds coefficients. An odds ratio of 1 suggests no impact.

The posterior estimates for the match-level random intercepts (α_{match}) reveal substantial variability in baseline log-odds of winning a match across matches. The estimated standard deviation, $\sigma_{\text{match}} = 3.03$ (95% credible interval: 2.49–3.63), indicates that some matches strongly favor the U.S. team, while others strongly favor the opponent. For example, matches with $\alpha_{\text{match}} \approx 3$ correspond to a predicted probability of winning an end of over 95% ($\frac{e^3}{1+e^3} \approx 0.95$), whereas matches with $\alpha_{\text{match}} \approx -4.5$ correspond to probabilities around 1% ($\frac{e^{-4.5}}{1+e^{-4.5}} \approx 0.011$). This large separation, combined with an estimated $\sigma_{\text{match}} \approx 3.03$, suggests that match context plays a dominant role in determining outcomes, potentially overshadowing the effects of Power Play usage and shot success.

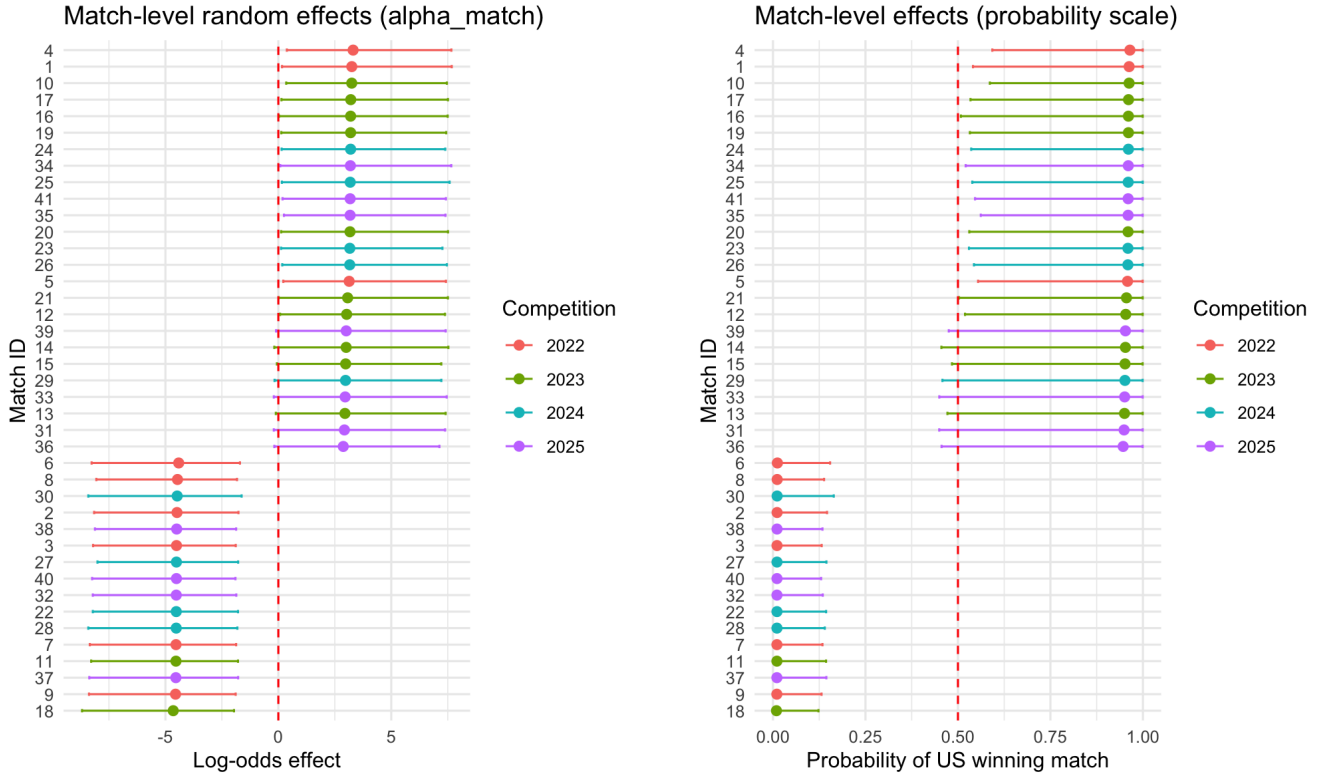


Figure 2: Posterior distribution of match-level random effects (α_{match}). Left: effects on the log-odds scale. Right: same effects transformed to probability scale via the logistic function.

The high effective sample sizes and $\hat{R}$ values of 1 for all α_{match} indicate that the estimates are robust and well-converged. Overall, the U.S. team's probability of winning an end is strongly conditioned on the specific match context, highlighting the importance of accounting for match-level variability in predicting performance.

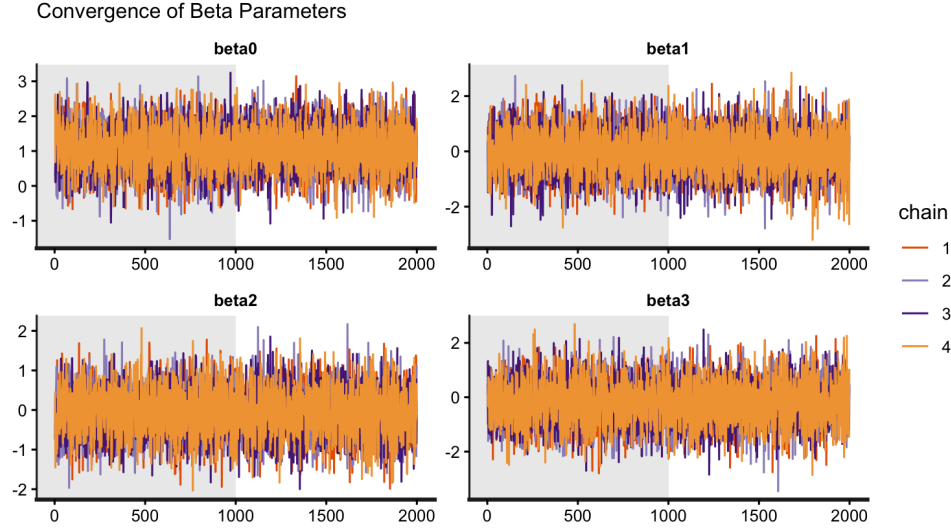


Figure 3: Convergence of Beta parameter Markov Chains

As shown in Figure [3], the traceplots for the four model coefficients show great mixing and convergence. Each plot resembles the gold standard for MCMC diagnostics. This indicates that the Monte Carlo algorithm has effectively explored the posterior distribution. We can see a clear shrinkage effect for matches where we have little data or where the results were extreme, the model doesn't just take those results at its face value. Instead, it pulls those estimates back toward the overall average of the entire dataset.

Discussion

- Do the odds of winning a match vary based on whether the team makes a power play during an end?

In short, choosing to use a Power Play doesn't actually change the overall odds of winning. As shown in Table 1, the coefficient for the Power Play ($\beta_1 = -0.04$) is nearly zero. Since the 95% credible interval on the log-odds scale widely overlaps zero, we have no evidence to suggest a strategic advantage. Converting this to the odds scale, the effect remains neutral (Odds Ratio ≈ 0.96). This implies that simply calling the Power Play doesn't boost for the U.S. team for winning. Other factors, such as the opponent's strength, carry more weight in determining the outcome.

- Do these results vary by whether or not the play (power or otherwise) "failed in its intended result"?

While it seems intuitive that making a shot should lead to bigger probability of winning, our findings shows the reality is more complicated.

1. Our variable β_2 measure whether a shot is successful bases on the player's intension. Even the execution is high quality, it doesn't have much to do with winning the final matches.

2. The interaction term β_3 , we looked into its odd ratio. It measures if a successful shot provides an extra boost specifically because it happened during a Power Play. Since β_3 is near zero (-0.23), there is no evidence showing that there is an extra boost. A successful shot is not more decisive during a Power Play to winning the match than it is in a standard end.

3. This anti-intuition situation which Power Play lack of success boost is likely due to a selection bias. US team often use a Power Play when they are already falling behind in the game. In these circumstances, even a perfect shot may not be able to overcome the opponent's lead. Therefore, our results don't suggest

that playing well is useless, but rather that the Power Play is often more like a signal that the team is already in a difficult position.

- Is it worth it to attempt to make a power play?

Yes, we think it is worth it as a last-minute tool. While the Power Play does not directly turn a losing end into a winning end, it still gives the game more possibilities. For a team that is already behind, a Power Play is the only way to create the chance to stay in the match. Therefore, it is worth it to try while already losing, but it is not worth it when the team is doing well.

- What impact, if any, did clustering or grouping have on your results?

Using Partial Pooling has a big impact on our result. In our model, the match-level intercepts α_{match} are drawn from the distribution $N(0, \sigma_{match})$. This model shrinks the matches with very few ends or highly unusual results toward the global average. This prevents a single outlier match biasing our coefficients disproportionately.

Our results revealed a high group-level standard deviation ($\sigma_{match} \approx 3.03$), which shows that the difficulty level varies drastically between matches. If we use a standard logistic regression without pooling, the model would have struggled to distinguish between a successful Power Play and a weak opponent. By using Partial Pooling, the model filters out noise from this match-level. It tells us that the Power Play's contribution to the winning is negligible when accounting how much harder it is to win against a strong team versus a weaker team.

Furthermore, the Partial Pooling Method addresses the small sample size issue within individual matches. Some matches in our dataset have only a few recorded ends. A No Pooling model would overfit to these matches, which Complete Pooling model would ignore their contexts. Partial Pooling makes to the best balance, providing a more trustworthy estimate of the Power Play's impact on the overall match outcome.

Stan Code

```
data {
  int<lower=1> N; // number of ends
  int<lower=1> M; // number of matches
  int<lower=1,upper=M> match_id[N]; // match for each end
  int<lower=0,upper=1> USWin[N]; // outcome: US wins?
  int<lower=0,upper=1> PowerPlay[N]; // yes/no powerplay
  int<lower=0,upper=1> Success[N]; // yes/no successful shot
}

parameters {
  real beta0;
  real beta1;
  real beta2;
  real beta3;
  vector[M] alpha_match; // random intercepts for matches
  real<lower=0> sigma_match; // SD of match intercepts
}

transformed parameters {
  vector[N] logit_p;
  for (i in 1:N) {
    logit_p[i] = beta0
                  + beta1 * PowerPlay[i]
                  + beta2 * Success[i]
```

```

        + beta3 * PowerPlay[i] * Success[i]
        + alpha_match[match_id[i]];
    }
}

model {
  target += normal_lpdf(beta0 | 0.5, 1);
  target += normal_lpdf(beta1 | 0, 1);
  target += normal_lpdf(beta2 | 0, 1);
  target += normal_lpdf(beta3 | 0, 1);

  target += normal_lpdf(alpha_match | 0, sigma_match);
  target += normal_lpdf(sigma_match | 0, 0.5);

  // Likelihood
  target += bernoulli_logit_lpmf(USWin | logit_p);
}

```

R Code

```

---
title: "TR2Code"
format: html
---

All clean and nice for you Taylor! Enjoy! :)

```{r}
library(tidyverse)
library(rstan)
curling <- read_csv("curling.csv")
```

```{r}
curling <- curling %>%
 mutate(Success = as.numeric(Points >= 3)) # define "success"
```

```{r}
curling <- curling %>%
 mutate(match_unique_id = group_indices(., CompetitionName, MatchID)) %>% # create unique id for
 mutate(comp_id = as.integer(factor(CompetitionName))) # integer competition_ids
```

```{r}
curling_data <- list(
 N = nrow(curling), # 311
 M = length(unique(curling$match_unique_id)), # 41
 match_id = curling$match_unique_id,

```

```

 USWin = curling$USWin,
 PowerPlay = curling$PowerPlay,
 Success = curling$Success
)

fit_curling <- stan(
 file = "curling.stan",
 data = curling_data,
 chains = 4,
 warmup = 1000,
 iter = 2000
)
'''

'''{r}
fit_curling # look at stan output
'''

'''{r}
Fixed effect parameters
fixed_effects <- c("beta0", "beta1", "beta2", "beta3")

Traceplots for fixed effects
traceplot(fit_curling, pars = fixed_effects, inc_warmup = TRUE)
'''

'''{r}
posterior distributions
library(bayesplot)
posterior <- as.array(fit_curling)

betas <- c("beta0", "beta1", "beta2", "beta3")

mcmc_areas(# Density plots computed from posterior draws with all chains merged, with uncertainty
 posterior,
 pars = betas, # defined above so i can be lazy here and other places
 prob = 0.95 # 95% credible intervals
) +
 geom_vline(xintercept = 0, linetype = "dashed", color = "red", linewidth = 0.8) + # line at 0
 ggtitle("Posterior Distributions of Beta Coefficients") +
 theme_minimal()
'''

'''{r}
caterpillar plot showing alpha_match
alpha_mat <- rstan::extract(fit_curling)$alpha_match # extract posterior draws of match-level ran

match_lookup <- curling %>%
 distinct(match_unique_id, comp_id) %>% # only one row representing match_unique_id and comp_id
 mutate(match_id = match_unique_id, # make names match in match_lookup and alpha_summary

```



```

 shortname = case_when(
 comp_id == 1 ~ "2022",
 comp_id == 2 ~ "2023",
 comp_id == 3 ~ "2024",
 comp_id == 4 ~ "2025")) # shorter, more logical names for CompetitionName (the plot looked

alpha_summary <- data.frame(
 match_id = seq_len(ncol(alpha_mat)), # 1 thru # of matches (41)
 mean = colMeans(alpha_mat), # mean of posterior draws for each match
 lwr = apply(alpha_mat, 2, quantile, 0.025), # 2.5%
 upr = apply(alpha_mat, 2, quantile, 0.975)) %>% # 97.5%
 left_join(match_lookup, by = "match_id") # join this and what competition it was in

alpha_summary <- alpha_summary %>%
 arrange(mean) # lowest to highest mean posterior draw

logoddpplot <- ggplot(alpha_summary, aes(x = mean, y = reorder(match_id, mean), # order highest to
 color = shortname)) + # color by competition
 geom_point(size = 2) + # plot the mean point
 geom_errorbarh(aes(xmin = lwr, xmax = upr), width = 0.2) + # add 95% cred int lines outward from
 geom_vline(xintercept = 0, linetype = "dashed", color = "red") + # red vertical line at 0
 theme_minimal() +
 labs(
 title = "Match-level random effects (alpha_match)",
 x = "Log-odds effect",
 y = "Match ID",
 color = "Competition"
)

convert to probability scale
alpha_summary_prob <- alpha_summary %>%
 mutate(
 prob_mean = exp(mean) / (1 + exp(mean)),
 prob_lwr = exp(lwr) / (1 + exp(lwr)),
 prob_upr = exp(upr) / (1 + exp(upr))
)

probability scale
probplot <- ggplot(alpha_summary_prob, aes(x = prob_mean, y = reorder(match_id, prob_mean), color =
 geom_point(size = 2) +
 geom_errorbarh(aes(xmin = prob_lwr, xmax = prob_upr), width = 0.2) +
 geom_vline(xintercept = 0.5, linetype = "dashed", color = "red") + # red vertical line at 50% p
 theme_minimal() +
 labs(
 title = "Match-level effects (probability scale)",
 x = "Probability of US winning match",
 y = "Match ID",
 color = "Competition"
)
)

```